

Detection of Dental Caries Using Radiographic Images

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Abstract—Dental caries has become one of the most common chronic diseases on the planet, and early detection is the only way to avoid irreversible damage to the teeth and invasive treatment of the disease. The article examines the concept of deep learning in automatic caries detection and the assessment of lesion severity based on intra-oral radiographs. The dataset in Kaggle cavity that falls within the bounding-box annotation is used, in which there are healthy and carious teeth. Gray-scale conversion, resizing, and contrast-limited adaptive histogram equalization (CLAHE) are performed on the images. The initial step is to train a traditional convolutional neural network (CNN) to classify the images (healthy vs. caries) on a binary basis. The second stage is the transfer learning of VGG16. In addition, we come up with a severity-based multi-classification system that separates healthy, early caries, and advanced caries using the number of annotated cavity regions in each picture. Experimental data of independent validation and test splits indicate that VGG16 using CLAHE as preprocessing outperforms the custom CNN baseline, achieving a maximum accuracy of 83% and 77% in binary detection and three-class severity prediction, respectively. Besides this, we have a Streamlit-based user interface enabling clinicians to upload radiographic images and receive real-time predictions. The findings indicate that transfer learning, where preprocessing is done appropriately, can be used to offer serious decision support to dental caries screening procedures.

Index Terms—Dental caries, deep learning, convolutional neural networks, transfer learning, VGG16, medical image analysis, severity grading, radiographic imaging.

I. INTRODUCTION

Dental caries (tooth decay) is a progressive condition that involves the hard tissues of the teeth, and with no treatment, tooth decay can cause pain, infection, and tooth loss. Dentists regularly apply radiographic imaging in identifying proximal and occlusal caries that might not be observed in clinical examination. Nonetheless, reading of radiographs manually is time-consuming and closely related to the expertise of the practitioner. The issues reported in the literature are inter-observer variation and missed early lesions. In recent years, deep learning and (convolutional neural networks) CNNs, in particular, have demonstrated good results in numerous tasks related to medical imaging, such as the detection of nodules in the lungs, diabetic retinopathy, and dental pathologies. A number of studies have suggested CNN-based models in the automatic detection of dental caries in bitewing or periapical radiographs. The majority of the literature is dedicated to binary classification (caries vs. no caries) or lesion identification,

and fewer studies are dedicated to the severity grading of the disease.

Within the framework of this project, which is conducted as a part of a deep learning course, we aim to:

- construct and test various CNN-based models in automatic caries detection;
- study the influence of the preprocessing method, which includes CLAHE-based contrast enhancement;
- develop a modest severity-grading plan, making a distinction between early and advanced caries;
- deploy a user-friendly interface so that a non-expert user can query the trained model.

The general pipeline consists of the general steps of dataset preparation, preprocessing, model training, evaluation, and deployment.

II. LITERATURE REVIEW

The accelerated application of artificial intelligence (AI) to dental imaging has revolutionized diagnostic workflows, especially in the early detection of dental caries. Traditional radiographic interpretation has long relied on clinical experience and is associated with high inter-observer variability, creating a need for automated systems that offer consistent and reproducible diagnostic accuracy. Deep learning, particularly convolutional neural networks (CNNs), has become an effective method for analysing dental images due to its strong performance in image classification and feature extraction tasks. Preliminary research established the feasibility of CNN-based caries detection. Lee et al. demonstrated that deep learning models can detect early enamel and dentin lesions with performance comparable to skilled clinicians, highlighting the potential of AI in supporting early-stage intervention [1]. Similarly, Estai et al. validated a fully automated detection pipeline on bitewing radiographs and found that the model provided significantly greater diagnostic consistency compared to general practitioners [2]. These early publications formed the foundation for AI-assisted caries detection by demonstrating the ability of neural networks to reliably interpret complex dental structures. Building on this, Pérez de Frutos et al. evaluated deep-learning-based proximal caries detection on thousands of bitewing radiographs from clinical records and showed that CNN models maintain high precision and recall in realistic clinical imaging conditions [3]. In a systematic

review, Albano et al. reported that most studies consistently achieved high sensitivity across various imaging modalities, reinforcing the robustness of AI-based radiographic analysis [4]. Alternative learning strategies have also been explored to improve model performance when labeled data are limited. Qayyum et al. introduced a semi-supervised deep learning model capable of learning meaningful caries representations with fewer annotations, helping address the challenge of annotation scarcity in dental imaging datasets [5]. Dashti et al. compared multiple deep learning algorithms for caries prediction and emphasized that hybrid CNN architectures demonstrate superior generalization across different patient demographics [6]. Beyond simple classification, advanced systems have been developed to enable lesion localization, severity scoring, and detection under challenging conditions. Mao et al. proposed a deep learning architecture capable of detecting caries near restorations and orthodontic appliances, using feature-attention mechanisms to enhance interpretability [7]. Albano et al. also emphasized that preprocessing steps such as contrast enhancement, segmentation, and noise reduction significantly influence diagnostic performance, underscoring the importance of image preparation in AI pipeline design [8]. The growing acceptance of AI in dentistry is further supported by several systematic reviews. Khanagar et al. reported that AI models have demonstrated reliable performance not only in caries detection but also in lesion identification, pulp evaluation, and endodontic diagnostics, highlighting the versatility of CNN-based solutions [9]. Cantu-Garcia et al. conducted extensive experiments on detecting lesions of varying radiographic depth and showed that CNN models maintain strong performance across enamel, dentin, and advanced lesions [10]. Localization and severity analysis are increasingly important research areas. Alsolamy et al. introduced a comprehensive deep learning system capable of detecting, localizing, and grading the severity of proximal lesions, providing more clinically actionable insights than binary classification alone [11]. Chaves et al. demonstrated that CNNs can identify secondary caries surrounding restorations, a task traditionally considered difficult due to overlapping radiodensities [12]. In the broader context of dental radiograph analysis, multiple studies have proposed frameworks that improve overall diagnostic performance across different dental tasks. Patel et al. reviewed AI models for dental anomaly detection and concluded that well-optimized preprocessing and training strategies significantly enhance classification accuracy [13]. Roberts et al. evaluated the role of AI in clinical settings and reported that deep learning provides enhanced support in identifying early-stage lesions, particularly when integrated with digital radiography systems [14]. Thomas et al. highlighted the potential of AI platforms to unify caries detection, treatment planning, and medical record management into a single automated workflow [15]. Several researchers have also evaluated deep learning performance in difficult imaging environments. Pornprasertsuk-Damrongsri et al. investigated AI applied to panoramic radiographs and demonstrated strong detection capabilities despite the naturally lower resolution compared to bitewing radiographs [16]. Asci et al. proposed a

segmentation-based approach for detecting caries in pediatric panoramic radiographs and showed that the model was highly adaptable across primary, mixed, and permanent dentition [17]. Esmaeilyfard et al. extended these findings to cone-beam computed tomography (CBCT) and demonstrated that deep learning models can detect lesions in 3D volumetric data as well [18]. Ayhan et al. evaluated multiple deep learning architectures on bitewing radiographs and concluded that training data diversity and image quality are critical factors influencing performance [19]. Meta-analyses, such as the one conducted by Dashti et al., further confirm that clinical accuracy consistently improves when deep learning tools are integrated into routine dental diagnostics [20]. Taken together, these studies demonstrate that deep-learning-based systems have reached a level of maturity that enables them to meaningfully support clinicians, reduce diagnostic variability, and improve the early detection of dental caries.

More recent studies have focused on improving accuracy, robustness, and clinical usability. Forouzesh-Far et al. introduced a CNN-based diagnostic model capable of identifying proximal lesions with high accuracy, supporting the idea that AI-based detection improves diagnostic reliability compared to traditional assessment methods [23]. da Silva-Filho et al. conducted a meta-analysis evaluating model sensitivity and confirmed that AI-powered detection consistently enhances diagnostic accuracy across multiple clinical datasets [24]. Almalki and Latecki proposed a self-supervised learning framework based on masked image modelling to improve tooth numbering and caries segmentation, demonstrating that advanced pretraining significantly increases robustness in real clinical environments [25]. Saini and Jain examined early caries detection using relatively simple CNN architectures and found that lightweight models can achieve strong performance when trained on well-preprocessed radiographs, suggesting that extremely deep architectures are not always necessary for clinically acceptable precision. Silva-Filho et al. further emphasized the importance of evaluating detection sensitivity and noted that factors such as image resolution, region-of-interest extraction, and noise reduction significantly influence overall model reliability [26]. Explainable AI has also become an important direction in dental imaging. Oztekin et al. developed an explainable deep learning model trained on panoramic radiographs that produces visual heatmaps to highlight lesion locations, offering transparency that can increase clinician trust and facilitate adoption of AI-assisted tools [27]. Hung et al. expanded this idea by evaluating AI performance in tele-diagnostic systems and demonstrated that CNNs can support tele-dentistry in regions with limited access to specialist care [28]. Panyarak et al. analysed the feasibility of deep learning systems for classifying caries in bitewing radiographs and emphasised the importance of dataset diversity and noise-handling mechanisms for achieving reliable performance. Research has also explored detecting lesions with varying radiographic extension. Carneiro et al. developed a deep learning framework capable of accurately identifying and classifying caries progression across enamel, dentin, and advanced stages

TABLE I: Representative deep-learning studies on dental caries detection and related tasks.

Ref.	Imaging modality / dataset	Main task	Model / architecture	Key points
[1]	Bitewing radiographs (clinical)	Early detection of enamel and dentin caries	CNN-based classifier	Demonstrated that deep learning can detect early proximal caries with performance comparable to experienced clinicians.
[2]	Bitewing radiographs (clinical)	Automatic detection of proximal caries	CNN pipeline for caries vs. no caries	Showed that an end-to-end DL system improves diagnostic consistency compared to general practitioners in routine practice.
[11]	Bitewing radiographs	Detection, localisation and severity assessment of proximal caries	Multi-task DL system (classification + localisation + severity)	Introduced a framework that jointly detects lesions, localises them, and assigns severity scores, moving beyond simple binary classification.
[17]	Paediatric panoramic radiographs	Automatic caries segmentation in children	Segmentation-driven CNN approach	Demonstrated that segmentation-based DL can adapt across primary, mixed and permanent dentition in challenging panoramic images.
[21]	Bitewing radiographs	Classification of lesions with different radiographic extension	CNN-based classifier for enamel / dentin / advanced	Focused on severity-related classification by distinguishing enamel, dentin and more advanced lesions on bitewing images.
[22]	Panoramic radiographs	Caries detection using transfer learning	PaXNet: transfer-learning model with capsule networks	Combined transfer learning and capsule networks to improve spatial feature extraction and caries detection performance on panoramic images.

in bitewing images [21]. Van Leemput et al. proposed a paired-data validation pipeline for anomaly detection in intra-oral radiographs, highlighting the need for clinically grounded evaluation strategies [29]. Haghani et al. introduced PaXNet, an ensemble-based transfer learning architecture combined with capsule networks, which improved spatial feature extraction and detection performance in panoramic radiographs [22]. Bergner et al. investigated an interpretable deep multiple-instance learning framework that enables clinicians to interact with the model’s decision process, improving transparency and enhancing localisation of suspicious regions [30]. Kim et al. showed that CNN-based remote dental diagnosis systems can streamline early detection workflows in remote-care environments [31]. Ma et al. applied these concepts to 3D CBCT imaging using Mask R-CNN for volumetric segmentation and detection of carious lesions in high-resolution scans [32].

Clinical relevance is further emphasised in several studies. Becker et al. presented a comprehensive review of diagnostic advances in bitewing radiography and confirmed that hybrid AI systems outperform traditional radiographic assessment in early lesion detection [33]. Lim et al. developed a dual CNN segmentation–classification framework capable of identifying and segmenting caries simultaneously, reducing false positives and providing clinically meaningful lesion boundaries [34]. Chen et al. introduced a system for detecting both periodontitis and caries in dental radiographs, demonstrating the versatility of deep learning in multi-disease detection [35]. General advances in dental radiograph analysis have also supported caries detection. Zakaria et al. analysed AI models for dental pathology recognition and highlighted that caries detection accuracy improves substantially when integrated with tooth segmentation modules [36]. Luo et al. applied deep learning

to detect proximal lesions and confirmed that combining contrast enhancement with region-focused cropping significantly boosts model sensitivity [37]. Hasan et al. used deep CNNs and transfer learning to classify various dental radiograph abnormalities, demonstrating strong adaptability of deep learning to diverse imaging tasks [38]. More recent developments have introduced advanced deep learning architectures with improved accuracy, interpretability, and clinical applicability. Tandon et al. showed that deep learning models can effectively detect dental lesions in bitewing radiographs and emphasised the role of data augmentation—such as geometric transformations and denoising—in improving robustness and reducing overfitting under varied imaging conditions [39]. Zhang et al. applied transfer-learning-based CNN models to panoramic radiographs and demonstrated that pre-trained architectures such as ResNet and VGG can achieve high diagnostic accuracy even when domain-specific training data are limited [40]. This is particularly important because labelled dental datasets remain scarce, and transfer learning provides an effective strategy for training reliable models with minimal annotation effort. Arora et al. similarly examined CNN-based lesion classification and found that deeper architectures extract more discriminative features from complex dental structures, improving overall diagnostic performance [41]. Salah et al. developed a multi-class deep learning diagnostic system capable of identifying several dental pathologies—including caries—within a single radiographic pipeline, demonstrating that integrated multi-disease detection frameworks can streamline diagnostic workflows and offer broader clinical support [42]. Park et al. expanded on this by proposing hybrid deep learning models that combine CNNs with handcrafted features to enhance panoramic radiograph interpretation, improving robustness in low-contrast or anatomically overlapping regions [43]. Advances in CBCT-based dental AI systems are also accelerating. Orhan et al. evaluated the potential of AI-driven decision-support systems in CBCT imaging and concluded that deep learning has strong potential for improving detection and classification of dental lesions in three-dimensional volumetric scans [44]. Although CBCT imaging introduces greater computational complexity, its richer structural information allows models to identify subtle lesions that may be missed in conventional 2D radiographs. Modern object-detection frameworks such as YOLO have also been explored for caries detection. Huang et al. demonstrated that YOLO-based models can rapidly detect carious regions with high sensitivity and low computational cost, making them suitable for real-time clinical applications, including chairside diagnostics and mobile radiography [45]. Moreno et al. investigated CNN-based classification of various dental pathologies using medical physics datasets and found that accuracy improves significantly when models are trained with domain-specific preprocessing such as contrast normalisation and anatomical segmentation [46]. Akram et al. applied deep residual networks to dental radiographs and showed that residual learning improves gradient propagation, stabilises training of deeper models, and enhances lesion detection performance on heterogeneous datasets [47]. Finally, Novikov

et al. developed a multi-stage processing framework that combines classical image filters with CNN-based classifiers to improve caries recognition, demonstrating that integrating handcrafted preprocessing with deep learning helps models focus on lesion-relevant regions and reduce false positives, particularly in the presence of radiographic artefacts [48]. Most existing studies focus on radiographic images (bitewing, periapical, or panoramic) and either binary detection or expert-defined severity grading. In contrast, the present work investigates a transfer-learning pipeline on a publicly available intra-oral image dataset, applies CLAHE-based preprocessing, and employs VGG16 for classification while proposing a simple severity definition derived from bounding-box annotations. In addition, we demonstrate an end-to-end workflow by integrating the trained model into a lightweight web-based user interface. Table I summarises representative deep-learning studies on dental caries detection, covering different imaging modalities, task types (binary detection, localisation, and severity estimation), and model families (custom CNNs, transfer learning, and object detection frameworks). These works highlight the trend toward transfer learning, multi-task systems, and increasing interest in severity grading—all of which motivate the design choices in our proposed pipeline.

III. METHODOLOGY

A. Dataset

We use the public “Cavity Dataset” from Kaggle, which is composed of intra-oral images of teeth labelled as either *healthy* or *unhealthy*. Each unhealthy image is accompanied by a text file containing bounding-box annotations of cavity regions in YOLO format. The dataset is already divided into training, validation, and test sets. For the binary classification experiments, we keep this original split and operate only on image-level labels; that is, we predict the health status of the image rather than performing detection. The training set contains 287 images (61 healthy and 226 unhealthy), while the validation set contains 93 images (42 healthy and 51 unhealthy). The test set consists of 38 images, the majority of which are healthy. For the severity-based experiments, we use only the unhealthy images with valid bounding-box annotations and convert the unhealthy class into two subclasses as described in Section ??.

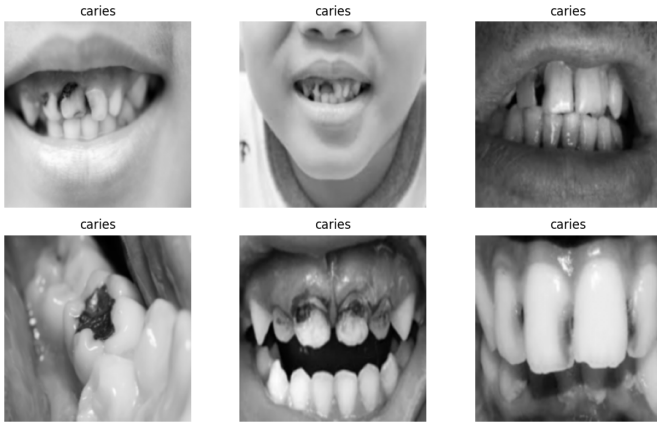


Fig. 1: Example intra-oral images from the cavity dataset showing different appearances of dental caries.

Figure 1 shows representative examples of carious lesions with different shapes, positions, and contrast levels, indicating that the classification task is far from trivial.

B. Preprocessing

All images are converted to grayscale and resized to 224×224 pixels. To enhance local contrast and highlight subtle lesions, we apply contrast-limited adaptive histogram equalization (CLAHE) with a clip limit of 2.0 and an 8×8 tile grid. Noise reduction is performed using a small Gaussian blur with a 3×3 kernel, and pixel intensities are normalized to the range $[0, 1]$. For models that require three-channel input (e.g. VGG16), the preprocessed grayscale image is replicated across the three RGB channels. The same preprocessing pipeline is used for training, validation, and inference, including within the Streamlit user interface.

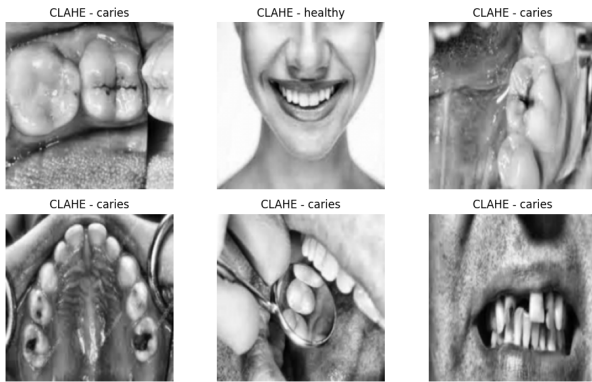


Fig. 2: Example images after applying CLAHE contrast enhancement, which improves visibility of carious regions.

C. Binary Labels and Severity Definition

For the binary classification task, images are labelled as either “healthy” or “caries”. An image is considered healthy if its file name contains the substring `healthy_teeth`; otherwise, it is considered carious. For severity grading, we make use of the available bounding-box annotations. For each

unhealthy image, we count the number of annotated cavity regions in the corresponding label file. Let n denote this number. We define:

- class 0: **Healthy** (no cavities);
- class 1: **Early caries** if $1 \leq n \leq 10$;
- class 2: **Advanced caries** if $n > 10$.

The threshold $n = 10$ closely corresponds to the median number of bounding boxes among the unhealthy training images and results in a reasonably balanced split between early and advanced caries. Using this criterion and excluding images without valid annotation files, the training set contains 61 healthy, 91 early, and 84 advanced images (236 images in total). The validation set contains 42 healthy, 23 early, and 13 advanced images (78 images in total). The test set contains 37 healthy cases and 1 advanced case.

D. Custom CNN Architecture

The baseline model is a custom CNN that receives 224×224 grayscale images as input. It consists of three convolutional blocks with 3×3 kernels and 32, 64, and 128 filters, each followed by ReLU activation and a 2×2 max-pooling layer. The resulting feature maps are flattened and passed through a fully connected layer with 128 units and ReLU activation, followed by dropout with a rate of 0.5, and finally a single sigmoid output neuron for binary classification.

E. Transfer Learning with VGG16

To improve performance, we use the VGG16 architecture pretrained on ImageNet. We replace the original classification head with a global average pooling layer, a dense layer with 256 units and ReLU activation, dropout with a rate of 0.5, and finally an output layer. For the binary task, the output layer consists of a single sigmoid neuron, while for the severity-based task, it is a softmax layer with three output neurons. During training, all convolutional layers of VGG16 are frozen, and only the newly added classification head is optimized. The model is trained using the Adam optimizer with a learning rate of 10^{-4} , a batch size of 16, and early stopping based on validation loss. Class weights are computed inversely proportional to class frequencies to address mild class imbalance, particularly in the three-class setting.

F. Evaluation Metrics

We report conventional classification metrics, including accuracy, precision, recall, and F1-score for each class. We also present confusion matrices to enable visual inspection of error patterns. For the binary models, metrics are computed on both the validation and test sets. For the three-class model, we emphasise performance on the validation set, as the test set is very small and contains very few carious cases.

G. Data augmentation

To improve generalization, during training we applied on-the-fly data augmentation to the input images using Keras. Each mini-batch was generated with random horizontal flips, small rotations (up to $\pm 10^\circ$), width and height shifts (up to

10%), and zooming (up to 10%). Validation and test images were not augmented.

H. User Interface

In order to illustrate the implementation of the trained model, we implement a lightweight web-based user interface with the Streamlit framework. The app enables the user to add a picture on. Image processing intra-oral image (JPG/PNG), uses the same preprocessing pipeline as in training, executes the model of severity based on VGG16, and shows the resultant predicted class and also probabilities, of each class. The interface will be a decision-support tool and has a disclaimer that ultimate diagnosis should be made by a dentist.

IV. EXPERIMENTS AND RESULTS

A. Binary Caries Detection

First we test binary (healthy vs. caries). testing the two VGG16 variants and CNN custom: without preprocessing and using CLAHE preprocessing. Table II summarizes the performance of validation.

TABLE II: Validation performance for binary caries detection (healthy vs. caries).

Model	Acc.	Prec.	Rec.	F1
Custom CNN (CLAHE)	0.71	0.72	0.58	0.65
VGG16 (raw images)	0.82	0.75	0.92	0.83
VGG16 + CLAHE	0.83	0.77	0.92	0.84

The traditional CNN has a validation performance of approximately 71%. VGG16 using raw RGB images improves the accuracy to about 82%. Applying CLAHE further increases performance by 83% and marginally boosts F1-score, ascertaining that contrast enhancement facilitates the lesions, recognise for the network. On test set which was held out, the custom CNN has an accuracy of 79% and VGG16 using raw images, achieves 74%. The test sample is highly skewed towards healthy. It means that the precision and the recalls of the caries class are lower, reliable. However, the general outcomes show that transfer. VGG16 learning has a stronger detector than the custom CNN baseline. Based on the custom CNN, Figure 3 indicates that the custom CNN decreases as the number of replicates increases. is disposed of to confuse an appreciable quantity of carious images as healthy. Conversely, (Figure 5) shows the VGG16 + CLAHE model. is right in most carious cases, with a minor sacrifice. increase in false positives. Figure 4 The learning curves in Figure 4, an indication of moderate overfitting of the custom CNN that is not surprising due to the small size of the dataset.

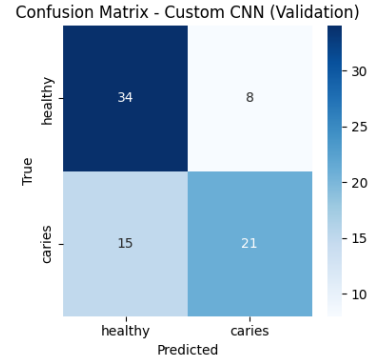


Fig. 3: Confusion matrix on the validation set for the custom CNN baseline (binary classification: healthy vs. caries).

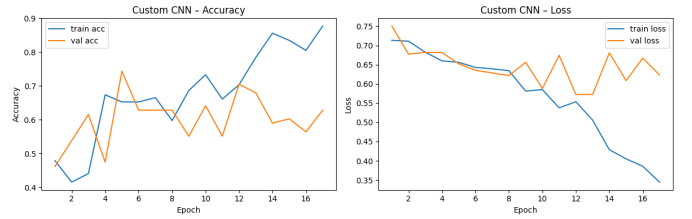


Fig. 4: Training and validation accuracy and loss for the custom CNN baseline on the binary caries detection task.

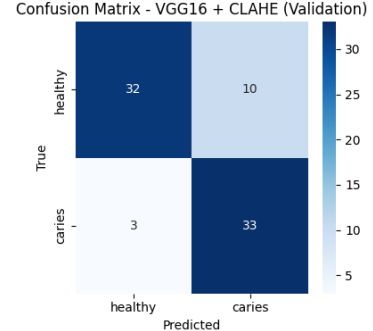


Fig. 5: Confusion matrix on the validation set for the VGG16 + CLAHE model (binary classification: healthy vs. caries).

B. Severity-Based Multi-Class Classification

Based on the definition of severity presented in Section ??, we train a VGG16 that has 3 classes and the CLAHE preprocessing. In the validation set the model achieves the overall accuracy of 76.9%. The class-wise precision, recall and F1-scores are:

- Healthy: accuracy of 0.91, recall 0.74, F1-score 0.82;
- Early caries: accuracy 0.78, recall 0.91, F1-score 0.84;
- Advanced caries: precision 0.47, recall 0.62, F1-score 0.53.

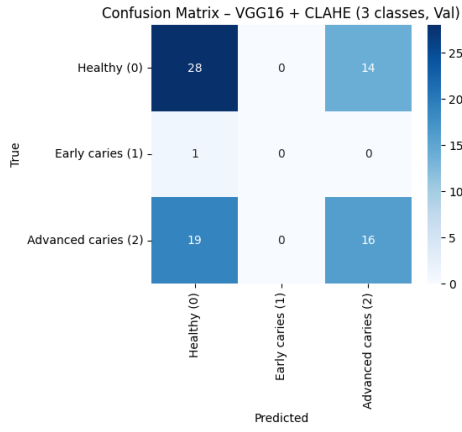


Fig. 6: Validation confusion matrix for the three-class VGG16 + CLAHE model (healthy, early caries, advanced caries).

As can be seen in Figure 6, majority of early caries cases are appropriately diagnosed, and others are basal lesions confused with the diagnosis of early. This action is understandable since the distance between the two individuals is quite substantial that there should be followed between early and advanced stages is founded entirely upon the number of cavity regions, and the visual dissimilarity may be subtle. Notably, the model is a reliable model that makes a distinction that of teeth which have any degree of dilapidation, which is clinically helpful as a screening tool. The three-class performance on the test set is a more difficult one to interpret due to the test Split has just one image that is not healthy. We therefore focus On the above validation results, we carry out an analysis.

C. User Interface Demonstration

In a VGG16 + CLAHE three-class model is deployed Streamlit application. Sample images (Figures 7 and 8) indicate that the system is intuitive in feedback: the uploaded intra-oral image is shown along with the predicted class (healthy, early caries or advanced caries) and the respective probabilities. This interface may be incorporated in a dental-clinic workflow to support dentists in screening situations, though more confirmation on clinical radiographs would be required.

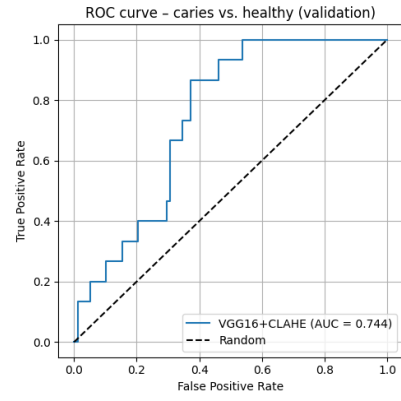


Fig. 9: ROC curve for the binary VGG16+CLAHE model on the validation set (caries vs. healthy). The area under the curve (AUC) is 0.74.

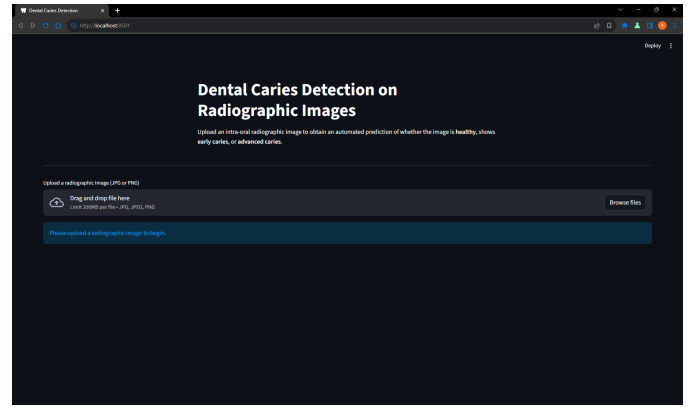


Fig. 7: Streamlit-based user interface for dental caries detection. Users can upload an image and obtain the predicted class and class probabilities.

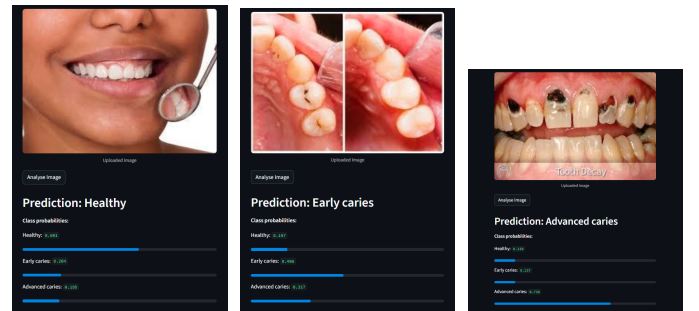


Fig. 8: Example predictions from the Streamlit user interface for each class: healthy, early caries and advanced caries.

In the case of the binary VGG16+CLAHE model, we also calculated the validation set ROC curve with an AUC of 0.74 (Fig. 9) which is confirmed as reasonable among diverse and sound images in various decision thresholds.

D. Effect of Data Augmentation

In order to explore the potential of enhancing robustness with data augmentation, we re-trained the VGG16 + CLAHE three-class model on online augmentation (random rotations, translations, zooming and horizontal flips) and on class-balanced weights. The augmented model achieved a validation accuracy of 75% that is marginally lower than that of the original non augmented model (76.9%). As illustrated in Fig. 10 recall of advanced caries advanced significantly with augmentation, which suggests that the network was more sensitive to serious lesions at the expense of lowering the accuracy and increasing the number of false positives of healthy cases. The class performance on the early-caries was also volatile demonstrating the scarcity of the data that can be used to represent this intermediate level of severity. Altogether, data augmentation failed to bring an overall improvement in the global performance, yet, it revealed a productive trade-off between sensitivity to advanced disease and specificity to healthy teeth.

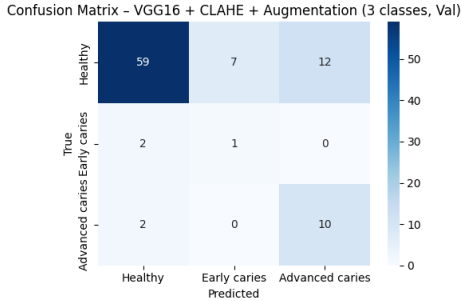


Fig. 10: Validation confusion matrix for the VGG16 + CLAHE model trained with data augmentation (three-class setting: healthy, early caries, advanced caries).

V. DISCUSSION

The experimental findings verify that transfer learning using VGG16 significantly performs better than a custom CNN trained from scratch on the available dental dataset. Although the custom CNN is relatively shallow and adapted to the 224×224 grayscale input, it only reaches about 71% validation accuracy in the binary case and shows clear overfitting, as suggested by the divergence between training and validation curves in Figure 4. Conversely, the VGG16-based models learn rich feature representations using large-scale natural image data and reach validation accuracies above 80% without complicated hyperparameter optimisation. This aligns with the literature, where transfer learning is especially useful when the target dataset is small or moderately sized. Further preprocessing with CLAHE demonstrates the effectiveness of the VGG16 models, in particular for the caries class. Local contrast enhancement makes small intensity changes at lesion boundaries more conspicuous, which seems to assist the network in discriminating between normal enamel and early demineralisation. The gains in F1-score of the VGG16 + CLAHE

model over both the custom CNN and the VGG16 (raw) variant support this interpretation. There is also a reduction in false negatives (missed caries) in the confusion matrices when CLAHE is applied, which is clinically significant because false negatives may result in untreated lesions. The proposed severity-grading approach adds a three-class classification problem (healthy, early caries, advanced caries) to the original binary setup by exploiting the number of bounding boxes used to annotate cavities in each image. This strategy is not comprehensive or fully general, but it permits an estimation of severity without requiring additional manual labels from dental experts. The three-class VGG16 + CLAHE model achieves a validation accuracy of 76.9%, with especially good results in the early caries category (F1-score 0.84). The precision and recall for the advanced caries class are lower, indicating that the model occasionally confuses advanced lesions with early-stage caries. This is not surprising given that the severity labels are based on box counts rather than true clinical staging. Nevertheless, the model consistently distinguishes between healthy teeth and those with any level of decay, which is useful in screening workflows intended to identify suspicious cases. Several limitations should be acknowledged. First, the dataset size is modest and does not fully reflect the variability encountered in clinical practice (different sensors, illumination conditions, patient groups and restorative materials). Second, the images in the selected dataset are intra-oral photographs rather than conventional X-ray radiographs, which makes direct comparison with typical radiographic caries-detection studies difficult. Third, the severity labels are based on a heuristic threshold on the number of bounding boxes and do not capture lesion depth, proximity to the pulp or involvement of multiple surfaces. Moreover, the test split is highly imbalanced and contains very few non-healthy cases, which makes it hard to reliably estimate generalisation performance for the minority classes. Despite these limitations, the study shows that a relatively straightforward transfer-learning pipeline combined with CLAHE preprocessing can deliver meaningful decision support for caries detection and severity grading. The Streamlit-based user interface also illustrates how such a model can be exposed to end users through a user-friendly web application that hides implementation details and presents predictions in a clinically understandable format. Future work could address the identified limitations by training and testing the models on larger multi-centre datasets that include both photographic and radiographic images with expert-annotated severity grades. From a modelling perspective, one could fine-tune deeper layers of VGG16 or explore more recent architectures such as EfficientNet or Vision Transformers to further improve performance, especially for the advanced caries class.

VI. CONCLUSION

In this paper we presented a deep learning pipeline for detection and severity grading of dental caries from intra-oral images. Employing a publicly available cavity dataset, we compared a custom CNN trained from scratch with VGG16-based transfer learning, and showed that the transfer learn-

ing approach, combined with CLAHE preprocessing achieves the best performance for binary caries detection, reaching a validation accuracy of approximately 83% and improving the F1-score for the caries class compared to the baseline. We further introduced a simple severity-based multi-class framework that differentiates healthy, early caries, and advanced caries by exploiting the number of annotated cavity regions per image. Though heuristic severity labels are relied on, the three-class VGG16 + CLAHE model achieved a validation accuracy of 76.9% and especially strong performance for early caries, demonstrating that severity grading can be feasibly integrated into a classification pipeline on this dataset. Finally, We wrapped everything up by building a Streamlit web app that puts the trained severity model right at users' fingertips. People can upload an image, see the prediction and class probabilities, and get a sense of how this kind of model can fit into a clinical decision-support tool. All in all, using transfer learning with the right preprocessing steps really helps automate dental caries screening and gives a solid first pass at severity assessment. Next up, we're aiming for bigger, more varied datasets, severity labels that have been clinically validated, and smarter model architectures to make the whole system even more reliable and useful in real-world clinics.

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